

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Uniform formatting throughout the project
2	Proper File Structure	Files and folders are logically organized	Yes	Organized folders for templates, static files and models.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Descriptive and easy-to-understand names.
5	Code Comments	Inline and block comments explain logic	Yes	Comments added in important code sections.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for training, prediction and UI.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Reusable logic with minimal duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Handles invalid inputs and runtime exceptions.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas	Training & Prediction	High
2	Model Training Module	Scikit-learn, XGBoost	ML Pipeline	High
3	Prediction Engine	Flask, Python	Loan Prediction	High
4	Dashboard & UI	HTML, CSS, JavaScript	Multiple Pages	High
5	Data Visualization Module	Matplotlib, Seaborn	Analytics Dashboard	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-structured modular architecture.
Readability	5	Clean and meaningful code.
Reusability	5	Reusable modules across the project.
Documentation / Comments	5	Adequate comments for key logic.
Overall Score	5	Well-organized and maintainable project.